

Dot plot

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Dot plot



One of the simplest and oldest methods to compare two protein or nucleotide sequences



It is a visual/graphic way of looking at regions of similarity between two sequences



How to do this?

- Two sequences of similar or variable length
- Write each letter of one sequence in a row
- Write each letter of the other sequence in column
- Start filling boxes where there is a letter for sequence 1 in 2 and for 2 in 1
- Interpret the plot



S1: GATTCTATCTAACT

S2: GTTCTATTCTAAC



	G	A	T	T	C	T	A	T	C	T	A	A	C	T
G	•													
T			•	•		•		•		•				•
T			•	•		•		•		•				•
C					•			•		•			•	
T			•	•		•		•		•				•
A		•					•			•		•		
T			•	•		•		•		•				•
T			•	•		•		•		•				•
C					•				•				•	
T			•	•		•		•		•				•
A		•					•			•		•		
A		•					•			•		•		
C					•				•				•	



	G	A	T	T	C	T	A	T	C	T	A	A	C	T
G														
T														
T														
C														
T														
A														
T														
T														
C														
T														
A														
A														
C														

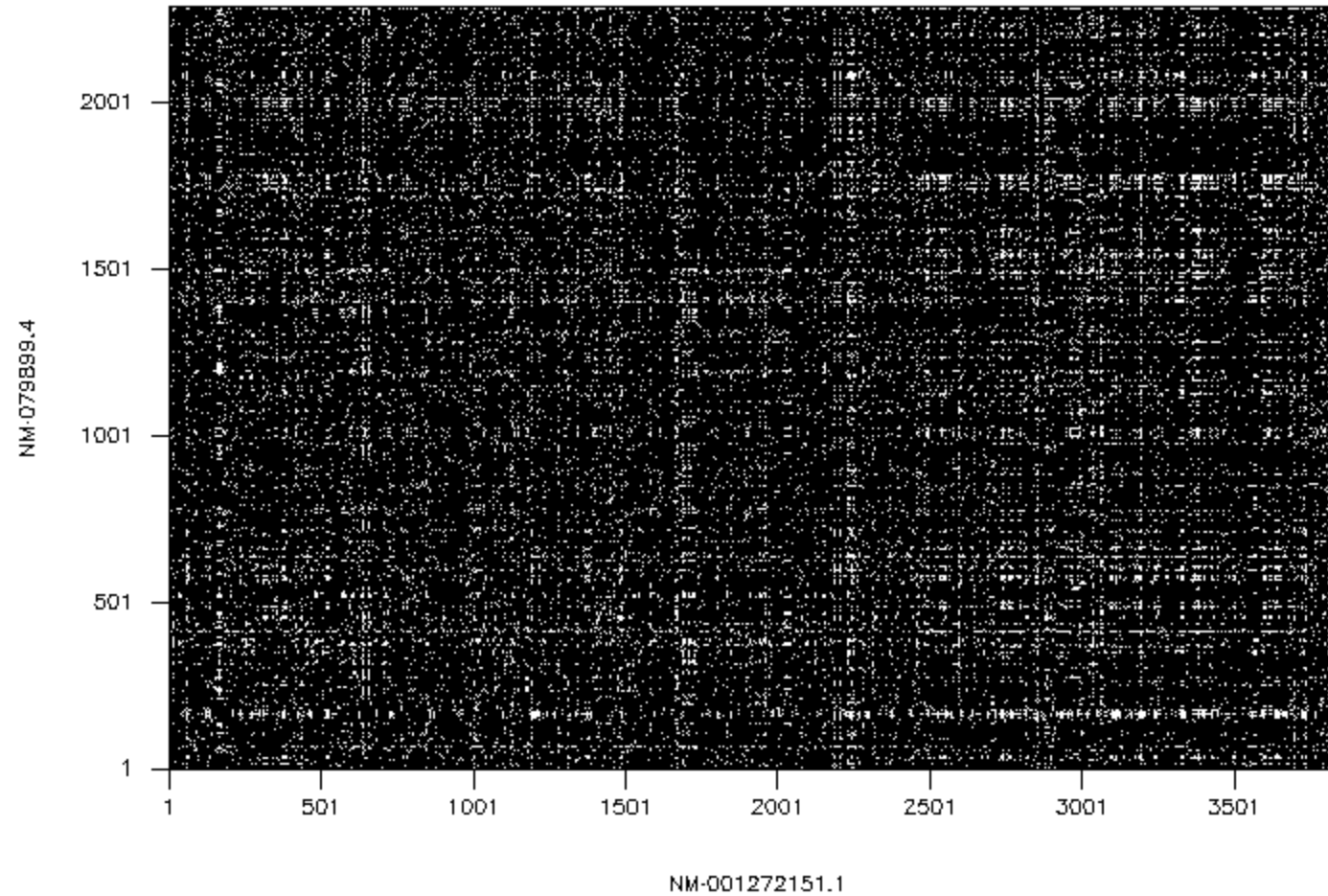


	G	A	T	T	C	T	A	T	C	T	A	A	C	T
G														
T														
T														
C														
T														
A														
T														
T														
C														
T														
A														
A														
C														



Dottup: fasta::emboss-dottup-l20201015-090214-0515-20194...

Thu 15 Oct 2020 09:02:18



D. melanogaster twin of eyeless (toy) transcript variant A and C: word size 1



Dot plots with defined word size and threshold

Word size = 1 (a)

DNA is made up of four nucleotide bases. There is a 25% chance that you will find a nucleotide match randomly by chance.

Word size = 3 (app)

appetizer, disappointed, application, appreciation, apprehensive, appreciative, approachable, inapplicable, misapplication, inappropriate, approximation, interlapping

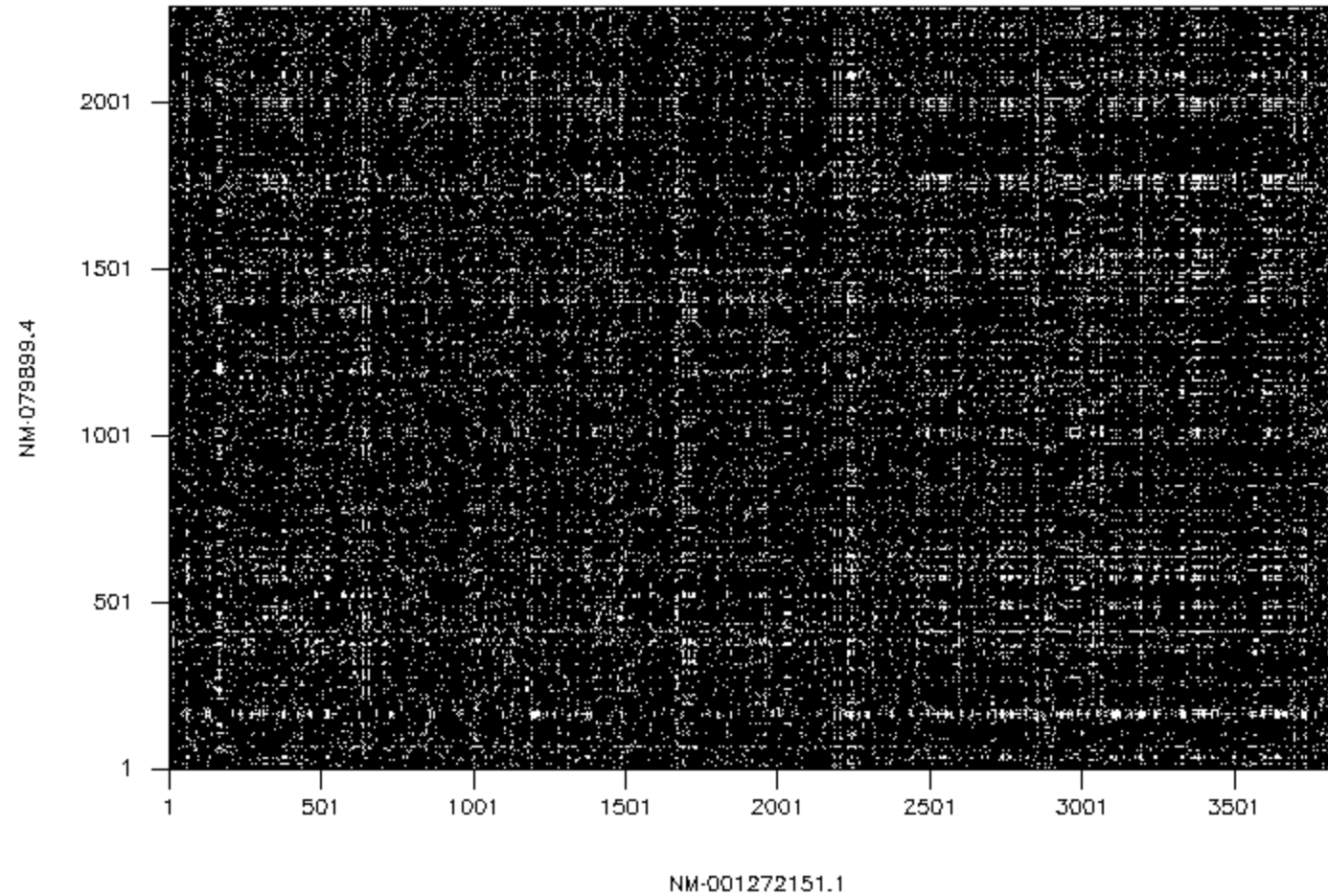
Word size = 5 (apple)

Applesauce, applejack, applecart, appleshare, appleworks, appletiser, applecross ...



Dottup: fasta::emboss-dottup-l20201015-090214-0515-20194...

Thu 15 Oct 2020 09:02:18

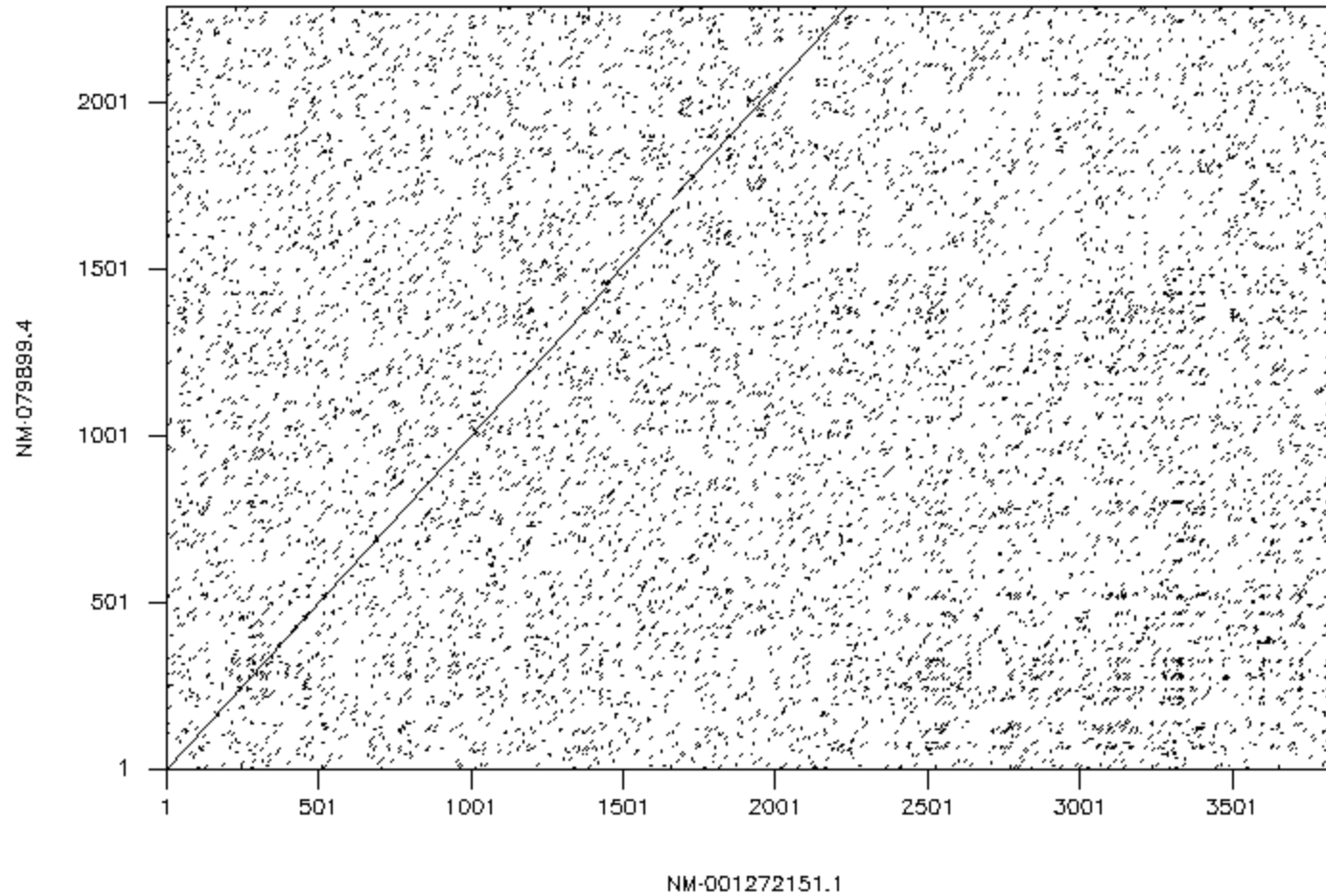


D. melanogaster twin of eyeless (toy) transcript variant A and C: word size 1



Dottup: fasta::emboss-dottup-l20201015-090345-0804-45019...

Thu 15 Oct 2020 09:03:48

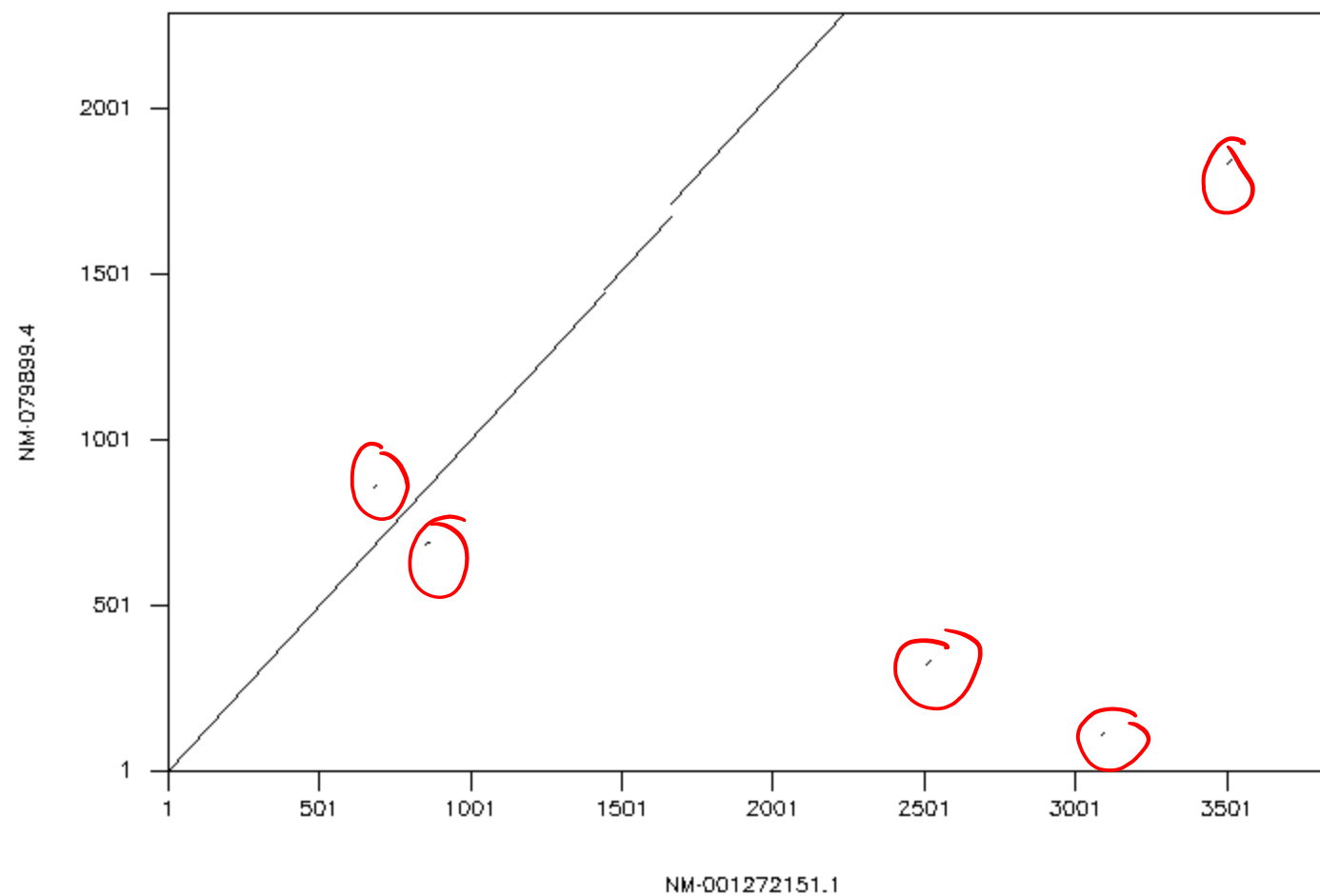


D. melanogaster twin of eyeless (toy) transcript variant A and C: word size 5



Dottup: fasta::emboss-dottup-l20201015-090423-0786-88092...

Thu 15 Oct 2020 09:04:26

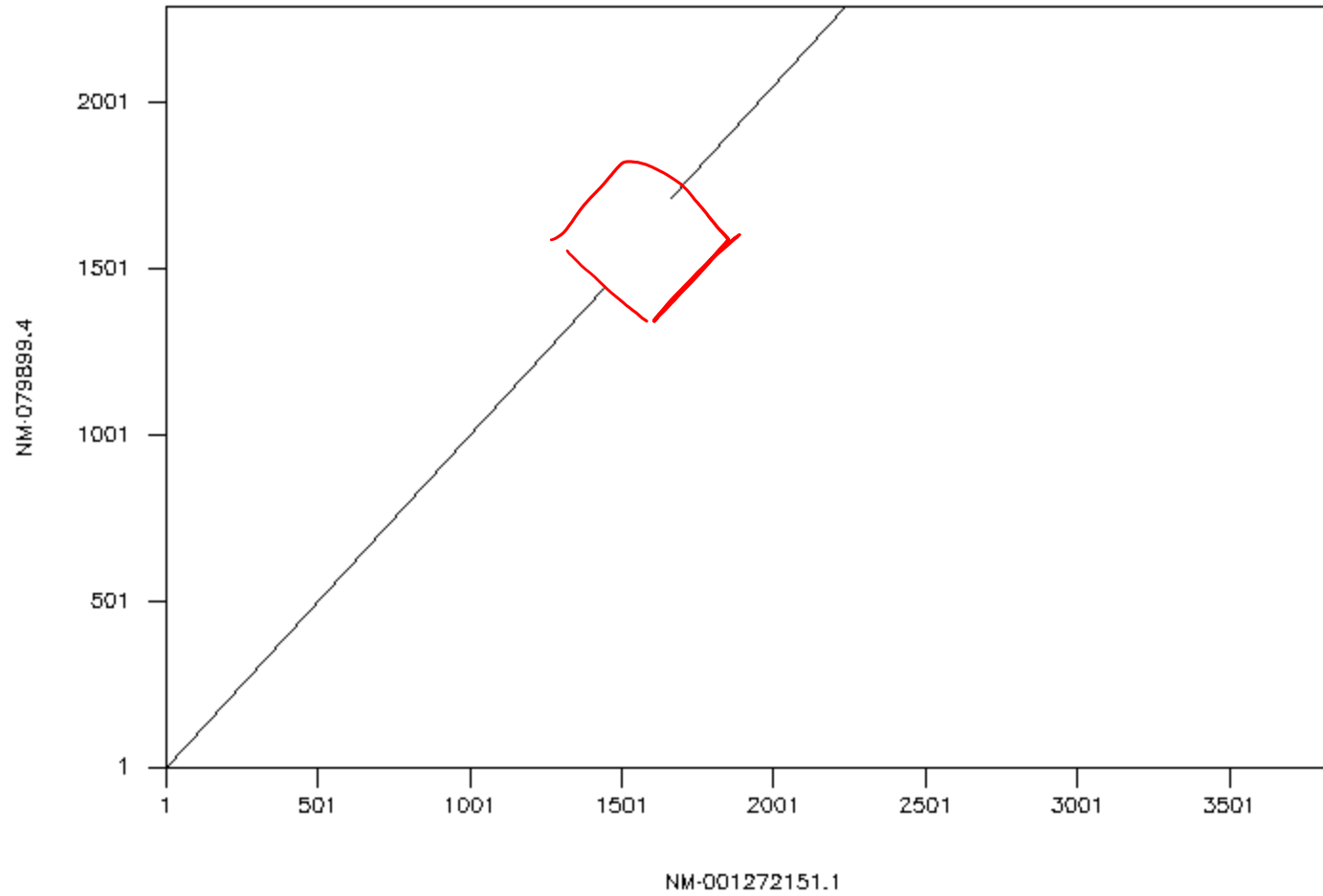


D. melanogaster twin of eyeless (toy) transcript variant A and C: word size 10



Dottup: fasta::emboss-dottup-l20201015-100621-0821-55382...

Thu 15 Oct 2020 10:07:00



D. melanogaster twin of eyeless (toy) transcript variant A and C: word size 250



Dot plots with thresholds

- If you colour in all cells with an identical letter, some dots may be due to **chance similarities**
- Therefore, it is common to use a **threshold** to decide whether to plot a ‘dot’ in a cell

A window of a certain size (eg. **window size** = 3) is moved up all possible diagonals, one-by-one

A score is calculated for each position of the window on a diagonal : the number of identical letters in the window

If the score is **equal to or above the threshold** (eg., threshold = score of 2), **all the cells in the window are coloured in**

The choice of values for the **window size** and **threshold** for the dot plot are chosen by trial-and-error



eg. for sequences "GCATCGGC" and "CCATCGCCATCG" , using a window size of 3, and a threshold of ≥ 2 :

	C	C	A	T	C	G	C	C	A	T	C	G
G												
C												
A												
T												
C												
G												
G												
C												

Score = 0, $\geq$ threshold $\rightarrow$ colour in

 = the sliding window

Example courtesy Dr. Avril Coghlan



	C	C	A	T	C	G	C	C	A	T	C	G
G												
C												
A												
T												
C												
G												
G												
C												

Word size = 1

	C	C	A	T	C	G	C	C	A	T	C	G
G												
C												
A												
T												
C												
G												
G												
C												

Word size = 3



Advantages

- Good for identification of long regions of strong similarity
- Easy to make and interpret
- Can be used for any length sequence

Disadvantages

- Need to find best window size
- Graphical representation doesn't give information about mutation



Seq#1: ATTGACAGTCGCCC

Seq#2: ATTGCCATTGCCATTGCC

